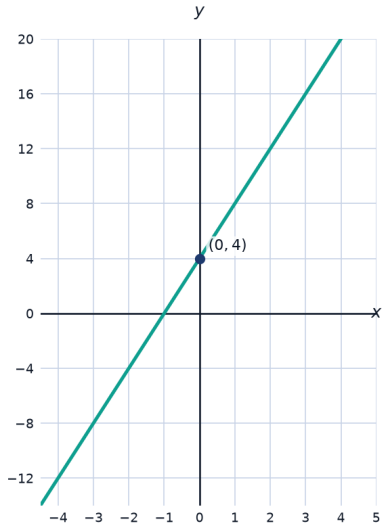
Polynomial Functions

ClipSAT Printable Practice Suite (clipsat.org) • Topic 3.01

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

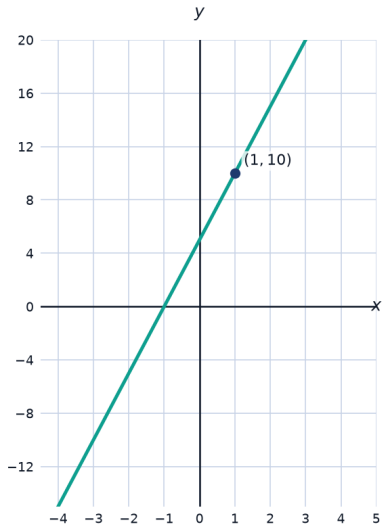
1 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

For the model $f(x) = 4x + 4$ in graphs and characteristics of polynomial functions, evaluate $f(5)$.



2 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

For the model $f(x) = 5x + 5$ in solving polynomial equations, evaluate $f(2)$.



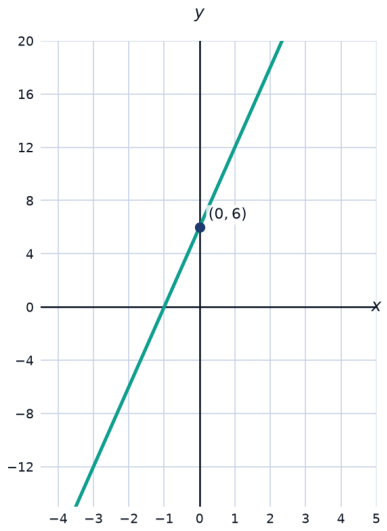
3 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A data set recorded for solving polynomial equations consists of the four values $\{15, 19, 23, 29\}$. What is the arithmetic mean? Show your complete algebraic work.

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4 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

For the model $f(x) = 6x + 6$ in remainder and factor theorems, evaluate $f(2)$.

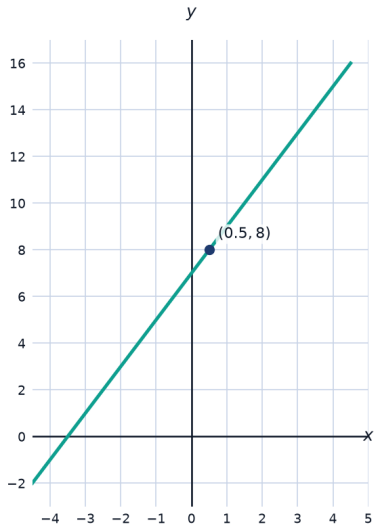


5 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A data set recorded for remainder and factor theorems consists of the four values $\{16, 20, 24, 30\}$. What is the arithmetic mean? Show your complete algebraic work.

6 [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

For the model $f(x) = 2x + 7$ in roots and zeros, evaluate $f(2)$.



7 [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

A data set recorded for roots and zeros consists of the four values $\{17, 21, 25, 31\}$. What is the arithmetic mean? Show your complete algebraic work.

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

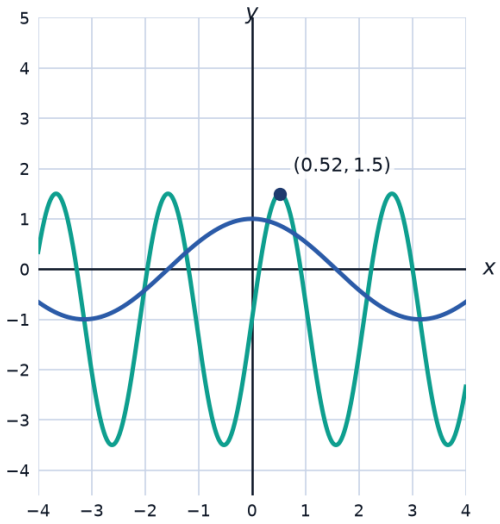
[Polynomial Functions · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

9 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Show that $(x - 2)$ is a factor of $p(x) = x^3 - 3x^2 + 4$, then fully factorise $p(x)$.

10 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

[Trigonometric Functions · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.



11 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

If $f(x) = x^2 - 1$ and $g(x) = \sqrt{x + 2}$, find (a) $f(g(3))$, (b) the domain of $g \circ f$.

12 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Level 1 · FoundationsLet $q(x) = -(x + 2)^2(x - 1)(x - 4)^3$. What is the degree of $q(x)$, does the graph cross or touch the x-axis at $x = -2$, and what is $\lim_{x \to \infty} q(x)$?

13 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

[Radicals & Rational Exponents · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

14 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

[Rational Expressions · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

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Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 8.01 Conic Sections

[Sequences & Series · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

16 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 9.01 Trigonometric Functions

Find the coefficient of x^2 in the expansion of $(3 - 2x)^5$.

17 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 1.01 Functions & Their Graphs

Find the value(s) of k for which $A = (k, 2; 3, k)$ is singular.

18 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 10.01 Matrices & Systems

Level 3 · AdvancedLet $R(x) = \frac{2x^2 - 8}{x^2 - 5x + 6}$. (a) Find the domain of $R(x)$. (b) Identify any holes and state their coordinates. (c) Write the equations of all vertical and horizontal asymptotes.

19 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 11.01 Probability & Statistics

Level 2 · MediumSolve the logarithmic equation $\log_2(x + 3) + \log_2(x - 1) = 5$ analytically for all real solutions. Show the verification step that eliminates extraneous roots.

20 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 2.01 Quadratics & Complex Numbers

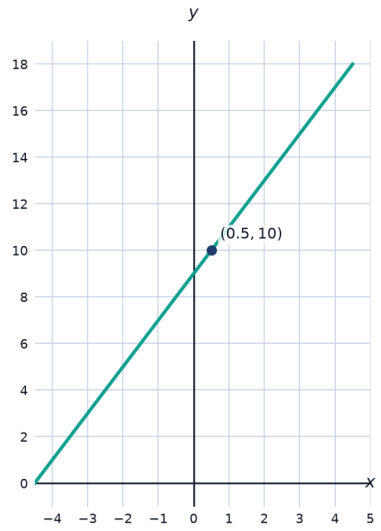
Level 2 · MediumThe function $f(x) = x^3 - 3x^2 - 9x + 2$ has a local maximum and a local minimum. Determine the intervals on which f is concave up and concave down, and find the coordinates of the inflection point.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 2x + 9$ in review: operations with polynomials, evaluate $f(2)$.



a. 13

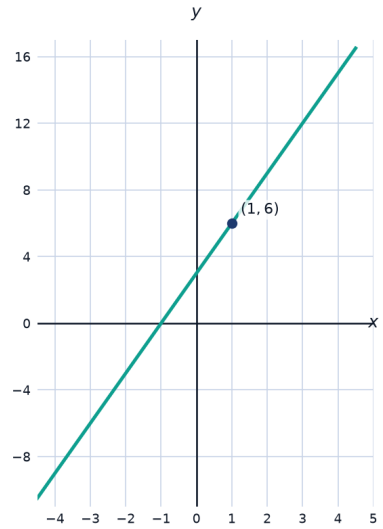
b. 15

c. 4

d. 17

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 3x + 3$ in synthetic and long division, evaluate $f(2)$.



a. 9

b. 12

c. 6

d. 13

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23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A data set recorded for review: operations with polynomials consists of the four values $\{12, 16, 20, 26\}$. What is the arithmetic mean?

a. 16.5

b. 17.5

c. 18.5

d. 20.5

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A data set recorded for synthetic and long division consists of the four values $\{13, 17, 21, 27\}$. What is the arithmetic mean?

a. 17.5

b. 18.5

c. 19.5

d. 21.5

25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

The Intermediate Value Theorem states that if $P(a)$ and $P(b)$ have opposite signs, then:

a. There are two roots

b. There is a root between a and b

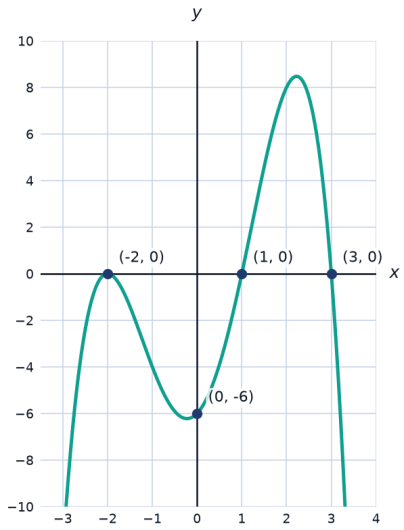
c. $P(0) = 0$

d. $P(a)P(b) = 0$

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26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

The graph of a polynomial function $f(x)$ of degree 4 is shown in the figure below. The graph is tangent to the x-axis at $(-2, 0)$, intersects the x-axis at $(1, 0)$ and $(3, 0)$, and crosses the y-axis at $(0, -6)$. A new polynomial $Q(x)$ is defined by $Q(x) = f(x) + \frac{1}{2}x^3$. When $Q(x)$ is divided by $x^2 - 4$, the remainder is the linear polynomial $R(x)$. What is the value of $R(5)$?



a. 24

b. 18

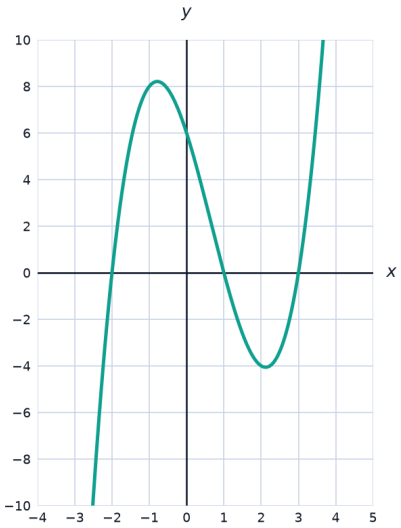
c. 17

d. 14

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27 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

The graph of a polynomial function $f(x)$ is shown below. Which of the following could be the equation for $f(x)$?



a. $f(x) = (x + 2)(x - 1)(x - 3)$

b. $f(x) = (x - 2)(x + 1)(x + 3)$

c. $f(x) = -(x + 2)(x - 1)(x - 3)$

d. $f(x) = (x + 2)(x - 1)^2(x - 3)$

28 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Which of the following statements describes the end behavior of the polynomial function $f(x) = -5x^4 + 2x^3 - 7x + 3$?

a. As $x \to -\infty$, $f(x) \to -\infty$, and as $x \to \infty$, $f(x) \to -\infty$

b. As $x \to -\infty$, $f(x) \to \infty$, and as $x \to \infty$, $f(x) \to -\infty$

c. As $x \to -\infty$, $f(x) \to -\infty$, and as $x \to \infty$, $f(x) \to \infty$

d. As $x \to -\infty$, $f(x) \to \infty$, and as $x \to \infty$, $f(x) \to \infty$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)**

Let $P(x)$ be a monic quartic polynomial with real coefficients whose graph is tangent to the x-axis at $x = -3$ and $x = 7$. A horizontal line $y = k$ intersects the graph of $y = P(x)$ at four distinct points whose x-coordinates, when listed in increasing order, form an arithmetic progression. What is the value of k?

a. 400

b. 225

c. 256

d. 625

30 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)**

Which of the following is a polynomial function?

a. $f(x) = \frac{1}{x}$

b. $g(x) = \sqrt{x}$

c. $h(x) = 3x^4 - 2x^2 + 5x - 1$

d. $k(x) = 2^x$

31 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

Show that the sum of zeros of a monic polynomial $x^n + a_{n-1}x^{n-1} + \cdots + a_0$ equals $-a_{n-1}$.

a. Cannot be determined

b. By Vieta: expanding $\prod(x - r_i)$, the x^{n-1} coefficient is $-\sum r_i = a_{n-1}$

c. Sum equals a_0

d. Sum equals n

32 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)**

Use the IVT to show $P(x) = x^3 - 2$ has a root between 1 and 2.

a. IVT does not apply

b. $P(1) = -10$, so by IVT a root exists in $(1, 2)$

c. $P(0) = 0$

d. $P(1) > 0$ and $P(2) > 0$, no root

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33 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

Find the cubic polynomial $P(x)$ with leading coefficient 2 whose graph passes through $(0, 6)$ and has zeros $1, -3$ and $1 + i$. But wait — to have real coefficients, if $1 + i$ is a root, so is $1 - i$. That gives degree 4.

a. $P(x) = 2x^3 - 6$

b. $P(x) = 2(x - 1)(x + 3)(x - 1 - i)$

c. $P(x) = 2(x - 1)^2(x + 3)$

d. This is impossible for a cubic with real coefficients (need degree ≥4 for those zeros)

34 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

Use synthetic division to show $(x + 1)$ divides $x^{100} - 1$.

a. Only for odd powers

b. $P(-1) = (-1)^{100} - 1 = 1 - 1 = 0$. Yes.

c. $P(-1) \neq 0$

d. Indeterminate

35 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)**

Use the Rational Root Theorem: list possible rational roots of $2x^3 - 3x + 1 = 0$.

a. $\pm 3, \pm \frac{1}{2}$

b. $\pm 1, \pm \frac{1}{2}$

c. ± 1

d. $\pm 1, \pm 2$

36 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)**

Write a cubic with roots $1, 2, 3$.

a. $x^3 - 6x^2 - 11x + 6$

b. $x^3 + 6x^2 + 11x + 6$

c. $(x - 1)(x - 2)(x - 3) = x^3 - 6x^2 + 11x - 6$

d. $(x + 1)(x + 2)(x + 3)$

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37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Apply Descartes' Rule of Signs to $P(x) = x^5 + x^4 - 3x^3 - 3x^2 + 2x + 2$.

a. At most 4 positive real roots

b. At most 1 positive

c. At most 2 positive real roots

d. Exactly 3 positive, exactly 2 negative

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 10.01 Matrices & Systems

The polynomial $P(x)$ has degree 4 and zeros $1, 1, 2, -3$. Write it in factored form (monic).

a. $(x+1)^2(x-2)(x+3)$

b. $(x-1)^2(x-2)(x+3)$

c. $(x-1)(x-2)(x+3)$

d. $(x-1)^2(x+2)(x-3)$

39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 11.01 Probability & Statistics

Find a degree-3 polynomial with zeros $0, -1, 4$ and leading coefficient 2.

a. $x(x-1)(x+4)$

b. $2x(x+1)(x-4) = 2x^3 - 6x^2 - 8x$

c. $2x^2(x-4)$

d. $2x(x-1)(x+4)$

40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 2.01 Quadratics & Complex Numbers

Find the quotient when $x^4 - 1$ is divided by $(x-1)$.

a. $x^3 + x^2 + x + 1$

b. $x^3 + 1$

c. $x^4 + 1$

d. $x^3 - x^2 + x - 1$

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41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 4.01 Rational Expressions

Find the value of k such that $(x-3)$ is a factor of $P(x)=x^3-kx^2+2x-3$.

a. $k=\frac{10}{9}$

b. $k=3$

c. $k=2$

d. $k=\frac{10}{3}$

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 5.01 Exponential & Logarithmic Functions

Determine: is $(x+2)$ a factor of $P(x)=x^3+x^2-4x-4$?

a. Only if degree is odd

b. No

c. Yes, $P(-2)=0$

d. Cannot determine

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 6.01 Radicals & Rational Exponents

Find the complex roots of $x^4-1=0$.

a. $x=\pm i$

b. $x=1,\ -1, i,\ -i$

c. $x=1, i$

d. $x=\pm 1$

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 7.01 Sequences & Series

How many turning points can a degree-n polynomial have (maximum)?

a. $n+1$

b. $n-1$

c. n

d. $2n$

45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 8.01 Conic Sections

Find all real zeros of $f(x)=x(x-2)(x+3)$.

a. $x=0,\ -2, 3$

b. $x=0, 2, 3$

c. $x=0, 2,\ -3$

d. $x=2,\ -3$

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